Then $\overrightarrow{AP}$ is perpendicular to $\vec{n}$.

So, $\overrightarrow{AP} \cdot \vec{n} = 0$ which gives $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$. ... (1)

which is the vector form of the equation of a plane passing through a point with position vector $\vec{a}$ and perpendicular to $\vec{n}$.

Note

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \Rightarrow \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} \Rightarrow \vec{r} \cdot \vec{n} = q, \text{ where } q = \vec{a} \cdot \vec{n}.$$

(b) Cartesian form of equation

If a, b, c are the direction ratios of $\vec{n}$, then we have $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$. Suppose, A is (x_1, y_1, z_1) then equation (1) becomes $((x - x_1)\hat{i} + (y - y_1)\hat{j} + (z - z_1)\hat{k}) \cdot (a\hat{i} + b\hat{j} + c\hat{k}) = 0$. That is,

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

which is the Cartesian equation of a plane, normal to a vector with direction ratios a, b, c and passing through a given point (x_1, y_1, z_1).

D5RXB1

6.8.3 Intercept form of the equation of a plane

Let the plane $\vec{r} \cdot \vec{n} = q$ meets the coordinate axes at A, B, C respectively such that the intercepts on the axes are $OA = a$, $OB = b$, $OC = c$. Now position vector of the point A is $a\hat{i}$. Since A lies on the given plane, we have $a\hat{i} \cdot \vec{n} = q$ which gives $\hat{i} \cdot \vec{n} = \frac{q}{a}$. Similarly, since the vectors $b\hat{j}$ and $c\hat{k}$ lie on the given plane, we have $\hat{j} \cdot \vec{n} = \frac{q}{b}$ and $\hat{k} \cdot \vec{n} = \frac{q}{c}$. Substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ in $\vec{r} \cdot \vec{n} = q$, we get $x\hat{i} \cdot \vec{n} + y\hat{j} \cdot \vec{n} + z\hat{k} \cdot \vec{n} = q$. So $x\left(\frac{q}{a}\right) + y\left(\frac{q}{b}\right) + z\left(\frac{q}{c}\right) = q$.

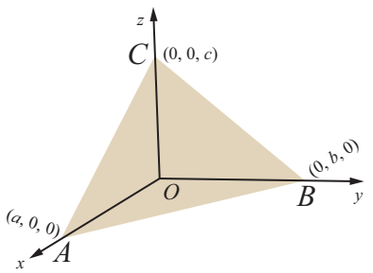


Fig. 6.26

Dividing by q, we get, $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. This is called the **intercept form** of equation of the plane having intercepts a, b, c on the x, y, z axes respectively.

Theorem 6.16

The general equation $ax + by + cz + d = 0$ of first degree in x, y, z represents a plane.

Proof

The equation $ax + by + cz + d = 0$ can be written in the vector form as follows

$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (a\hat{i} + b\hat{j} + c\hat{k}) = -d$ or $\vec{r} \cdot \vec{n} = -d$.

Since this is the vector form of the equation of a plane in standard form, the given equation $ax + by + cz + d = 0$ represents a plane. Here $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$ is a vector normal to the plane. ■

Note

In the general equation $ax + by + cz + d = 0$ of a plane, a, b, c are direction ratios of the normal to the plane.

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